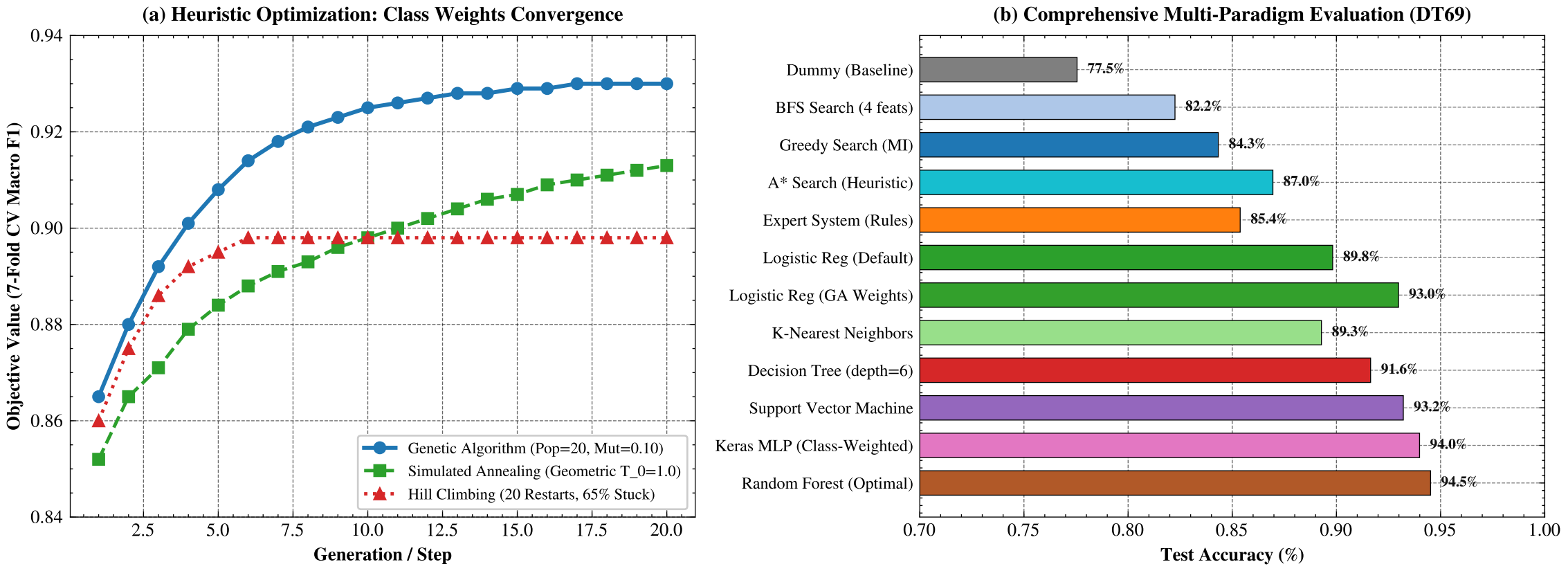
(a) Heuristic Optimization: Class Weights Convergence
Objective Value (7-Fold CV Macro F1)
Generation / Step
0.94
0.92
0.90
0.88
0.86
0.84
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Genetic Algorithm (Pop=20, Mut=0.10)
Simulated Annealing (Geometric T_0=1.0)
Hill Climbing (20 Restarts, 65% Stuck)
(b) Comprehensive Multi-Paradigm Evaluation (DT69)
Dummy (Baseline)
77.5%
BFS Search (4 feats)
82.2%
Greedy Search (MI)
84.3%
A* Search (Heuristic)
87.0%
Expert System (Rules)
85.4%
Logistic Reg (Default)
89.8%
Logistic Reg (GA Weights)
93.0%
K-Nearest Neighbors
89.3%
Decision Tree (depth=6)
91.6%
Support Vector Machine
93.2%
Keras MLP (Class-Weighted)
94.0%
Random Forest (Optimal)
94.5%
0.70
0.75
0.80
0.85
0.90
0.95
1.00
Test Accuracy (%)